Structured hybrid at m=3

Structured hybrid at m=3: mixed m=2 / 2+1 relations, cycle stop, a=0 n=37

Exploratory, local run (no AWS identity in this session; not an isolab-eks run). Not a leaderboard row (validation gate not met).

Question

Can feeding the cycle-stop echelon with a mixture of two-summand relations (source A) and three-summand “2+1” relations (source B, a fraction q of the base-point/sign trials) beat the pure two-summand cycle stop (q=0)? Curve: GF(2^37) Koblitz a=0 (cofactor 596), d=12, K=2011, 8 q values x 6 seeds, 48 runs, all verified (`ic_verified`, `rho_verified` true; bad_t = inconsistent = 0). Instrument checks: q=0 reproduces the merged m=2 trial counts 6/6; 96/96 small-n records verified.

Prediction (frozen before the sweep, PROTOCOL.md)

Calls to stop rise monotonically with q; no q beats q=0; q=1 is near pure m=3 (~94.7 M calls from #3212). **Outcome: confirmed on all three.**

Result (analysis.txt, ratios.csv)

q	mean calls	vs q=0 (geomean, 95% CI)	rels A / B	closed by
0	2.44 M	1	142 / 0	A 6/6
0.000083	3.26 M	1.33	142 / 48	A 6/6
0.00025	4.90 M	2.01	142 / 148	A 6/6
0.00075	9.80 M	4.02	142 / 442	A 6/6
0.0022	19.5 M	8.2 (6.1-9.8)	115 / 1032	A 4, B 2
0.008	24.6 M	10.6 (7.3-17.1)	43 / 1416	B 6/6
0.025	31.0 M	14.0 (10.1-21.9)	20 / 1821	B 6/6
1	94.7 M	42.9 (31.5-68.2)	2 / 5550	B 6/6

Monotone in q; no q is below 1. For q <= 0.00075 the stop is still closed by two-summand rows, so the ratio is a pure cost overhead of the extra trials (seed-independent, hence the degenerate intervals there). The mixed rows never made a short dependency appear earlier: the 3-term rows need ~0.9K relations to close (rank ~1850 at stop) while 2-term rows close at 142. q=1 reproduces pure m=3 (94.68 M vs 94.7 M).

Analytic side result (hybrid_estimate.csv)

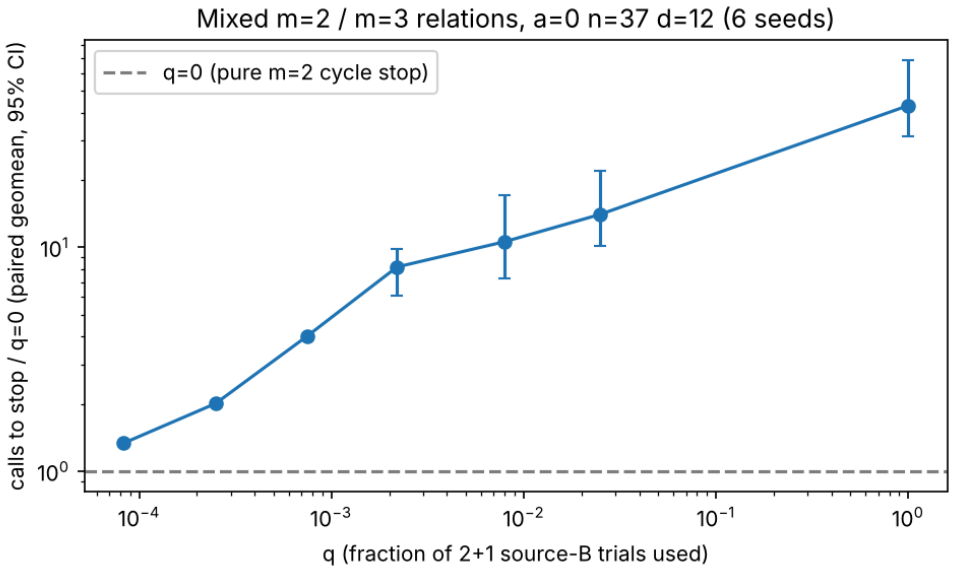
Hybrid XL (guess g bits, then XL on the descended S4 system) has cost 2^g * M^2 minimised at pure enumeration of V^3. Heuristic table, log2 cost at n=37: best hybrid 36.0, pure XL 69.6, 2+1 split 23.3. The 2+1 split is the cheapest m=3 route per relation, and it still loses at the stop.

Conclusion

No structured hybrid improves on the two-summand cycle stop at n=37; best measured algebraic variant stays q=0 (about 7,700x the matched rho at d=12, see earlier reports). This is a negative result, not an improvement; gap to rho unchanged.

Limits

6 seeds per q; only the 2+1 construction used as source B; mixing weights q not optimised beyond the grid; a=0 n=37 only; local run.



ratio